

Formulae

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Discrete

Uniform

- **Mean:** $\frac{a+b}{2}$
- **Variance:** $\frac{1}{12}(b-a)(b-a+2)$

Bernouli

- **Mean:** p
- **Variance:** $p(1-p)$

Binomial

- **Mean:** np
- **Variance:** $np(1-p)$

Geometric

- **Mean:** $\frac{1}{p}$
- **Variance:** $\frac{1-p}{p^2}$

Continuous

Uniform

- **Mean:** $\frac{a+b}{2}$
- **Variance:** $\frac{(b-a)^2}{12}$

Exponential

- **Mean:** $\frac{1}{\lambda}$
- **Variance:** $\frac{1}{\lambda^2}$
- **PDF:** $\lambda e^{-\lambda x}$
- **CDF:** $1 - e^{-\lambda x}$

Normal

- **Mean:** μ
- **Variance:** σ^2